Organic Cumulative Exam
September 21, 1991

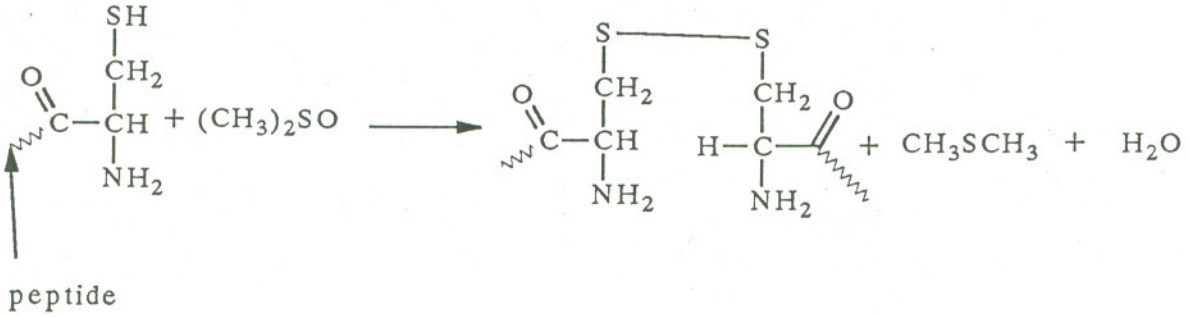
The following reactions are taken from recent articles in the *Journal of the American Chemical Society*.

(1) "Disulfide Bond Formation in Peptides by Dimethyl Sulfoxide. Scope and Applications", *113*, 6657.

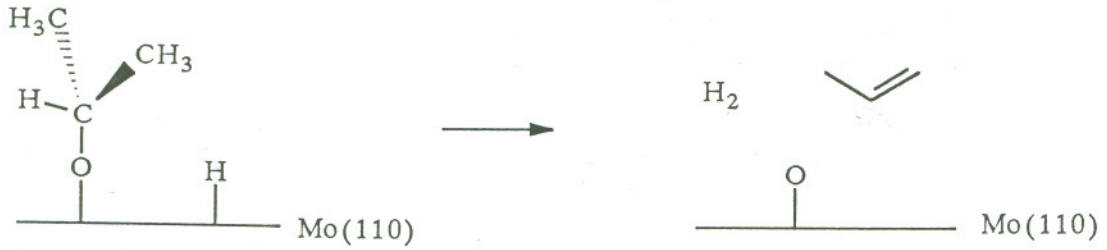
(2) "Selective γ C-H Bond Cleavage in Alkoxides: 2-Propanol on Mo(110)", *113*, 6686

(3) "Carbon-Carbon Bond Formation in Aqueous Ethanol: Diastereoselective Transformation of Unprotected Carbohydrates to Higher Carbon Sugars Using Allyl Bromide and Tin Metal", *113*, 6674.

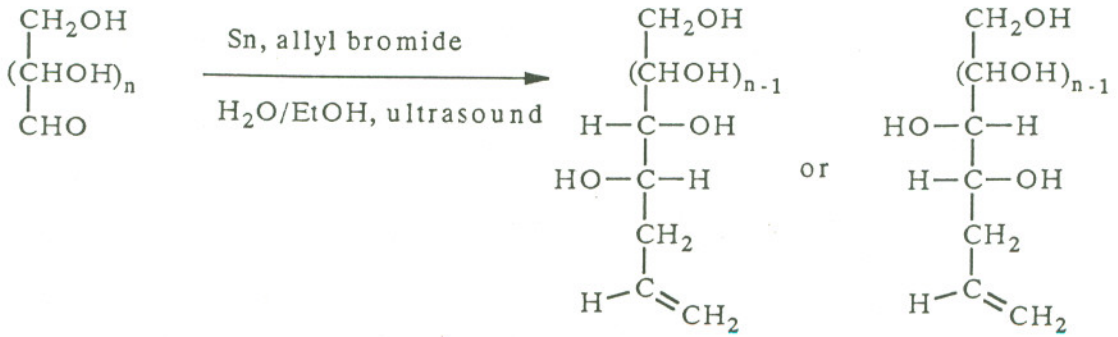
(1)



(2)



(3)



Answer the following questions:

1.1 Propose a reasonable mechanism for the transformation shown in (1) using curved arrow notation.

1.2 How is your mechanism consistent with the fact that the rate of disulfide bond formation is independent of pH between pH 3 and 8?

2.1Propose a reasonable mechanism for the transformation shown in (2) using curved arrow notation.

2.2 In general, the C-H bond dissociation energy for the C-H bond adjacent to the oxygen is 4-7 kcal/mole *weaker* than the methyl C-H bonds of 2-propanol, when bound to metal surfaces. Explain.

3.1Propose a reasonable mechanism for the transformation shown in (3) using curved arrow notation.

3.2 Assign configurations (*R* or *S*) to the Fischer projections of the products shown.